



# S130 PDK Release Notes

V2.1.302

April 20, 2022

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## S130 PDK Release Notes

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## Introduction

This document describes the enhancements made to each S130 Process Design Kit (PDK) released to date.

**Note:** Before contacting [SkyWater Technology](#) to report any new S130 PDK installation or user concerns, review the PDK support documentation for possible solutions. In the Cadence command interpreter window (CIW), select **SkyWater > PDK Documents** or view the `$s130/pdk/v2.1.302/DOC` directory.

## V2.1.302 (Current Release)

This PDK release features two updated documents, a new antenna diode, a new areaid marker layer, accumulated model changes, and numerous design rule checking changes, including changes to pattern density and latchup rules.

**Note:** SkyWater removed the `./IO` directory and contents from the S130 PDK. The S130 I/O cell IP is now available as a separate product download.

## Document Files

- Updated the *S130 Design Rule Manual*, Spec No. 002-22690.
- Updated the *S130 PDK User Guide*, Spec No. 002-22691.
- Removed the *S130 Topside Via Physical Design* application note; content moved to the *S130 Design Rule Manual*, Spec No. 002-22690.
- Removed the obsolete *Topological Design Rules User Guide*, Document No. SW-00-000002.

## Library Files

- Added the diode\_antenna pcell.
- Updated the `S130.layermap` file to include the areaid:lv\_net layer for marking latchup exemptions.
- Modified the Cadence Virtuoso ADE Explorer SkyWater menu to indicate that the Model graphical user interface (GUI) is not yet available for use; it will be available in a future PDK release.

## Model Files

### V1.1.0 (April 12, 2022)

- Enabled “mismatch\_factor” as a global mismatch multiplier.
- Enabled “process\_mc\_factor” as a global Monte Carlo process multiplier.
- Added eta0 variation to increase the NMOS field-effect transistor (FET) drain-induced barrier lowering (DIBL) effect to match worst-case leakage specs (Xena ticket 1008).

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- Added temperature coefficients to the 300  $\Omega$ /sq. poly resistor, rpoly\_hp.
- Reduced the 116 Å thick gate oxide matching constant to  $5.8 \times 10^{-3}$ , scaled from  $T_{ox} = 41$  Å.

### V1.0.1 (September 3, 2021)

- Added temperature coefficients to the 300  $\Omega$ /sq. poly resistor, rpoly\_hp.

### V1.0.0 (May 3, 2021)

- Released newly restructured corner, matching, and Monte Carlo process models for FETs, resistors, capacitors, diodes, bipolar junction transistors, and electrostatic discharge (ESD) protection devices.

## Physical Verification—DRC

- Added a required header to the Calibre design rule checking (DRC) files.
- Added layers areaid:lv\_net and r1c:drawing.
- Enabled pseudo-fill GDSII output stream files.
- Added switching that generates local pattern density results only when the GENERATE\_LOCAL\_DENSITY switch is turned on.
- Added two latchup checks that run during base DRC when the LATCHUP switch is turned off.
- Removed the seal-ring placement dependency from all latchup checks.
- Corrected the stress rule <metX>OverCA definition.
- Changed stress spacing checks to use the input metal.
- Changed the stress area calculation algorithm to prevent stress area calculation when an areaid:critCorner or areaid:critSid marker layer is not present (Xena ticket 1039).
- Modified the stress rules to improve fill layer recognition.
- Added the following DRC rules:
  - ndiff.latchup.1 and pdiff.latchup.1
  - nwell.CON.2, nwell.CON3 (Xena ticket 1072)
  - pwell.OVL.1 (Xena ticket 1107),
  - r1c.CON.1, r1c.ENC.1, r1c.LEN.1, r1c.SP.1, and r1c.WID.1
- Revised the following DRC rules:
  - latchup.generic.3b, latchup.generic.5a/b (Xena ticket 1162), latchup.misc.4, and latchup.signal.12a (Xena ticket 1162)
  - metX.chip.clear\_area.max\_DEN, metX.chip.clear\_area.min\_DEN, metX.local.clear\_area.max\_DEN, and metX.local.clear\_area.min\_DEN, where metX = met1–met5
  - metY.chip.oxide.low\_DEN.1 and metY.local.oxide.low\_DEN.1, where metY = met1–met4
  - nwell.CON.1 (Xena ticket 1072)
- Removed the following DRC rules:
  - metX.chip.high.DEN.4, metX.chip.low.DEN.3, metX.local.high.DEN.2, and metX.local.low.DEN.1, where metX = met1–met5

## Sample Files

- Added the `areaid:lv_net` marker layer to the `display.drf` file.
- Updated the `cds.lib` file to point to the S130 DIG and IO libraries that are available separately.

## Technology Files

- Added the `areaid:lv_net` marker layer to the `S130.tf` file.

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## V2.1.301 (December 8, 2021)

This PDK release features the new topside via. Use of this feature is currently restricted to specific customers; contact [SkyWater Technology](#) for more information.

### Document Files

- Added the *S130 Topside Via Physical Design* application note that describes the design layers and rules that support the new topside via.
- Updated the *S8 Technology Family Topological Design Rules* document, Spec No. 001-01905.

### Library Files

- Set the sealing pcell maximum die width and length parameters to 100,000  $\mu\text{m}$ .
- Added the viatop and cmtest drawing and mask layers to the stream layer map file.

### Program Files

- Renamed files to remove spaces and special characters.

### Physical Verification—DRC

- Added topside via design rules to the DRC deck.
- Added metal test design rules to the DRC deck for internal use only.

### Sample Files

- Added the viatop and cmtest drawing and mask layers to the `display.drf` file.

### Technology Files

- Added the viatop and cmtest drawing and mask layers to the tech file.

## V2.1.2 (September 30, 2021)

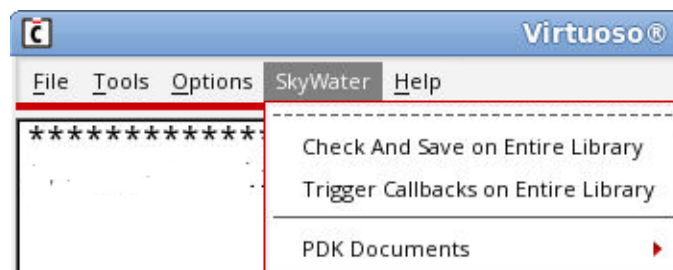
This PDK release featured new documentation, new continuous device models that replaced discrete device models, updated DRC and layout versus schematic (LVS) checking decks, and updated technology Library Exchange Format (LEF) files.

### Document Files

- Added the new *S130 PDK User Guide*, Spec. No. 002-22691, to assist users with the S130 process design kit, from installation to physical design verification and simulation.
- Replaced the *S130 (S8) Technology Family Design Manual*, Spec No. 002-21993, with an initial draft of the new *S130 Design Rule Manual*, Spec No. 002-22690. These design rules are synced with the updated S130 DRC deck and supersede all S130 design rules described in other PDK documents.

### Library Files

- Added a new SkyWater menu to the Cadence CIW to facilitate (a) Check and Save on Entire Library, (b) Trigger Callbacks on Entire Library, and (c) convenient access to PDK documentation, as shown in the following figure:



- Replaced the diff:hv layer used for 12 V and 20 V diffusion source and drain recognition with v12 and v20 marker layers.
- Renamed the ntap\_li and ntap\_met1 multipart paths (MPPs) to nw\_li and nw\_met1, respectively. Added new v5, v12, and v20 MPPs.
- Added the anchor cell.
- Removed the fuse cell and made the associated target layer invalid.
- Removed device size limitations that restricted parameters, such as width and length, to discrete values. Minimum and maximum values are retained.

To convert devices with discrete width and length parameters in existing designs to the continuous device models, type `load(strcat(getShellEnvVar("PDK_HOME") "/PROGS/SKILL/discrete2continuous.il"))` into the CIW. Run **CIW > SkyWater > Trigger Callbacks on Entire Library** after running the `discrete2continuous.il` program.

- Added editable *external* width and length parameters to the searing pcell. The internal width and length parameters used in existing designs can be viewed but not edited. SkyWater highly recommends using the updated searing pcell, if possible.

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## Physical Verification

### Design Rule Checking

**Note:** The current S130 fill deck requires a Siemens Calibre YieldEnhancer SmartFill license. SkyWater plans to include a basic fill deck in a future PDK release.

- Updated v5, v12, and v20 n-well design rules to use the highest voltage marker over nwell as the n-well voltage.
- Removed Rule v12.SP.1.
- Replaced the existing fuse design rules with two new checking-only rules that flag all fuse shapes as errors.
- Updated the bond pad design rules to remove exemptions.
- Updated the latchup design rules to reduce false checking errors.

### LVS Checking

- Added the keyword “inner” to several HOLE commands to prevent LVS checking errors caused by deep n-well diode placement.

## Technology Files

- Updated the `s130.tlef` file to support abstract generation.
- Updated the `s130.tlef` file to prevent Cadence Innovus from requiring a cut layer above met5.
- Corrected the enclosure values in the via sections of the `s130.tlef` file.



## **V2.0.0 (November 15, 2020)**

This release was the first full external release of the S130 PDK, which supports the S8 process with overhauled PDK collateral: rewritten device parameterized cells (pcells), expanded electronic design automation (EDA) tool support, and simplified physical verification rules.

**Note:** The SkyWater S130 semiconductor manufacturing process technology is not backwards compatible with its S8 predecessor process technology.